

L4

CLASS X - SCIENCE

METALS AND NON-METALS

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Check yourself

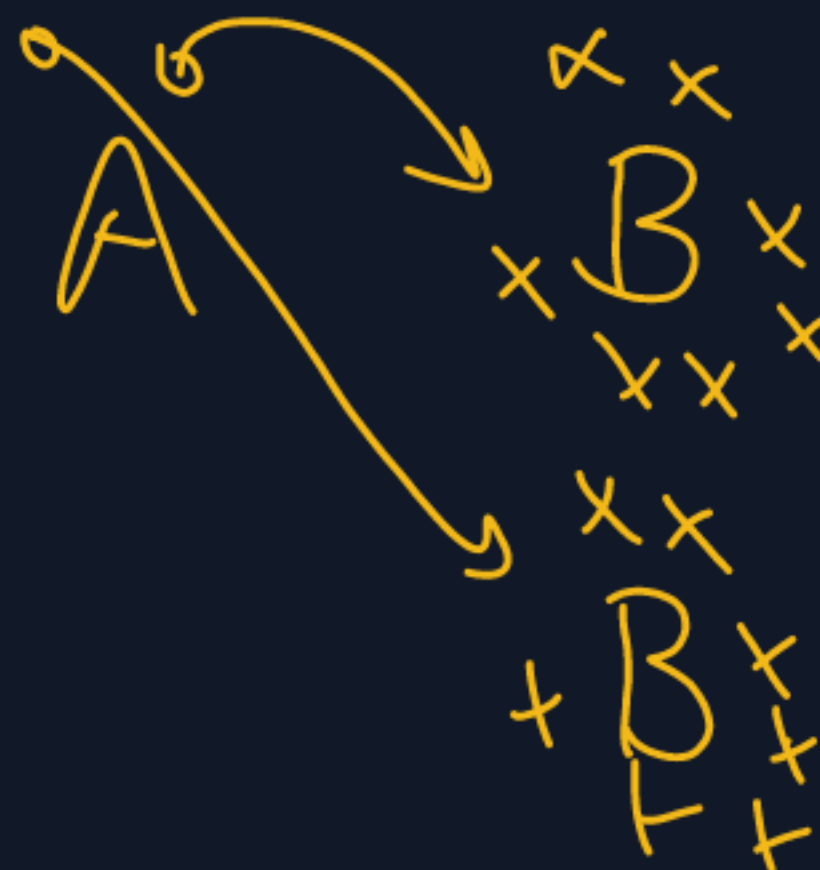
Ques. Consider elements A (2,8,2), B (2,7), C (2,8,1), and D (2,8,8).

(i) Which is most reactive metal?

(ii) Which is noble gas?

(iii) Write the formula of compound formed between A and B.

(iv) Which bond will it be?



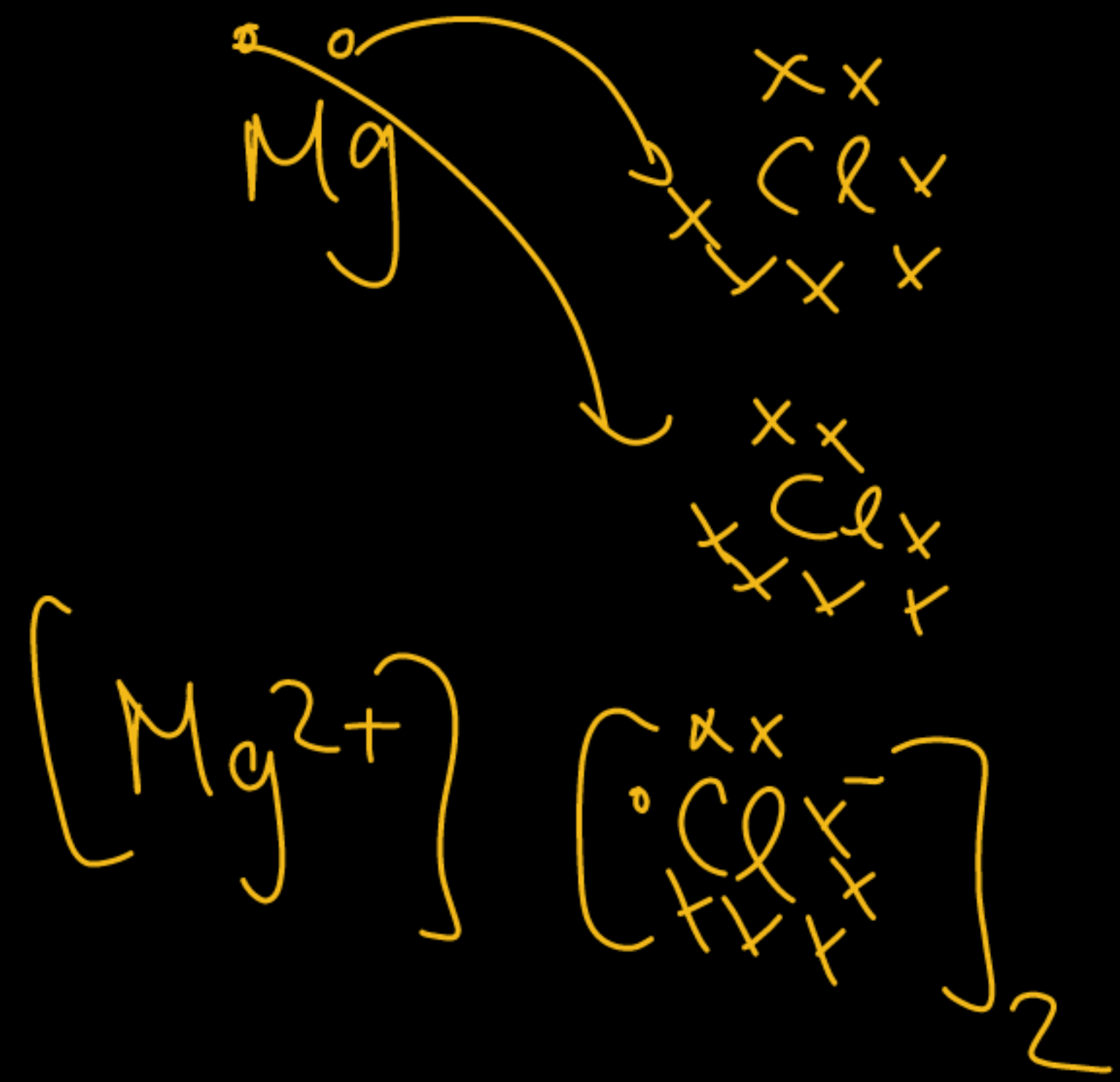
Check yourself

QUES: A student explains the formation of Magnesium chloride as follows:

- ~~1. Magnesium gains two electrons.~~
- ~~2. Chlorine loses one electron.~~
- ~~3. MgCl_2 is formed because opposite charges attract.~~
- ~~4. Solid MgCl_2 Conducts electricity because it contains ions.~~

Identify all incorrect statements, rewrite them correctly, and explain how electron transfer leads to ionic bond formation.





Check yourself

QUES: A student tested the electrical conductivity of different substances using a bulb and recorded the following observations:

Condition	Observation
Solid NaCl	Bulb does not glow
Molten NaCl	Bulb glows
NaCl solution	Bulb glows
Sugar solution	Bulb does not glow

Answer the following:

- Why does solid NaCl not conduct electricity?
- Why do molten NaCl and aqueous NaCl solution conduct electricity?
- Why does sugar solution not conduct electricity even though sugar dissolves in water?
- What conclusion can be drawn about the electrical conductivity of ionic compounds?

Check yourself

QUES: An engineer has to choose one material for making the inner lining of a high temperature furnace.

Two materials are available:

Material (A) : Ionic compound

Material (B) Covalent molecular compound

Which material should be selected? justify.

Answer: **Material A should be selected.**

Ionic compounds have strong electrostatic forces of attraction between oppositely charged ions, giving them high melting points and making them suitable for high temperature applications.



Check yourself

QUES: Assertion (A) : A laboratory technician could identify an unknown sample as an ionic compound because it conducted electricity only after being melted. (T)

Reason (R): The mobility of ions depends on the physical state of the ionic compound.

- ☒ a) Both A and R are true, R is the correct explanation of A.
- b) Both A and R are true, R is not the correct explanation of A.
- c) A is true, R is false.
- d) A is false but R is true.



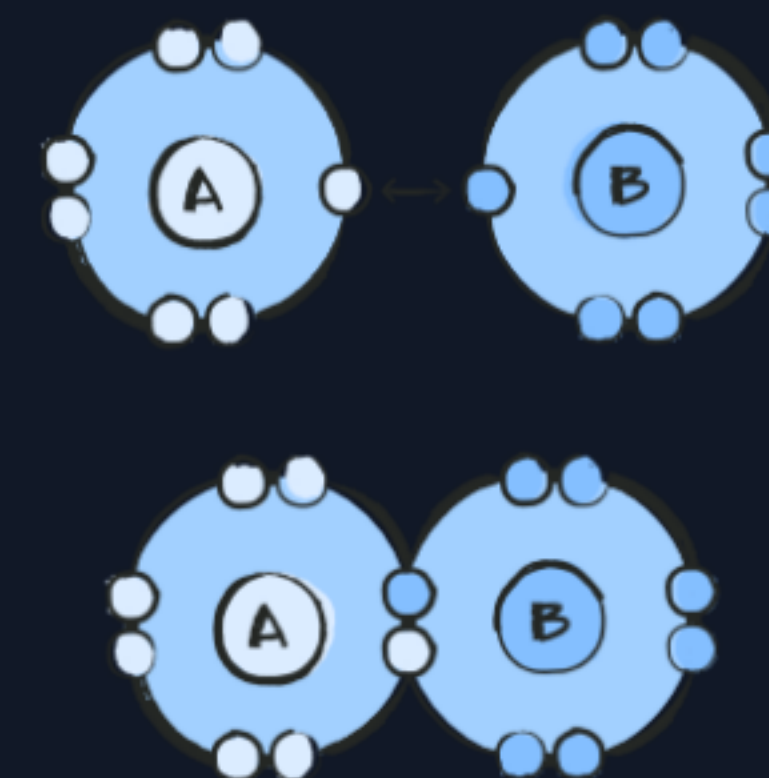
Properties of Ionic Compounds

1. Physical Nature:

- They are solids and hard due to strong force of attraction between oppositely charged ions.
- Generally brittle → break into pieces when pressure is applied.

2. Melting & Boiling Points: High melting and boiling points → large energy needed to break strong inter-ionic forces.

3. Solubility: Generally soluble in water. Insoluble in organic solvents (petrol, kerosene, etc.).



NaCl
↓
"Namak"

4. Conduction of Electricity:

→ cation
→ anion

- Conduction requires movement of charged particles (ions).
- Solid state: Do not conduct electricity (ions fixed in rigid structure).
- Molten state: Conduct electricity (electrostatic forces broken by heat → ions move freely).
- Aqueous solution: Conducts electricity (ions dissociate and move to opposite electrodes).



How do atoms become stable?

features	✓ metals	non metals
valence electrons	usually 1-3	usually 5-7
<u>tendency</u>	lose electrons easily	gain electrons easily
<u>reasons</u>	easier to lose a few electrons than to gain many.	need only a few electrons to complete the shell
<u>ion formed</u>	positive ion (cation,+)	negative ion (anion)
example	Na (2,8,1) - $\text{Na}^+ + \text{e}^-$	$\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$

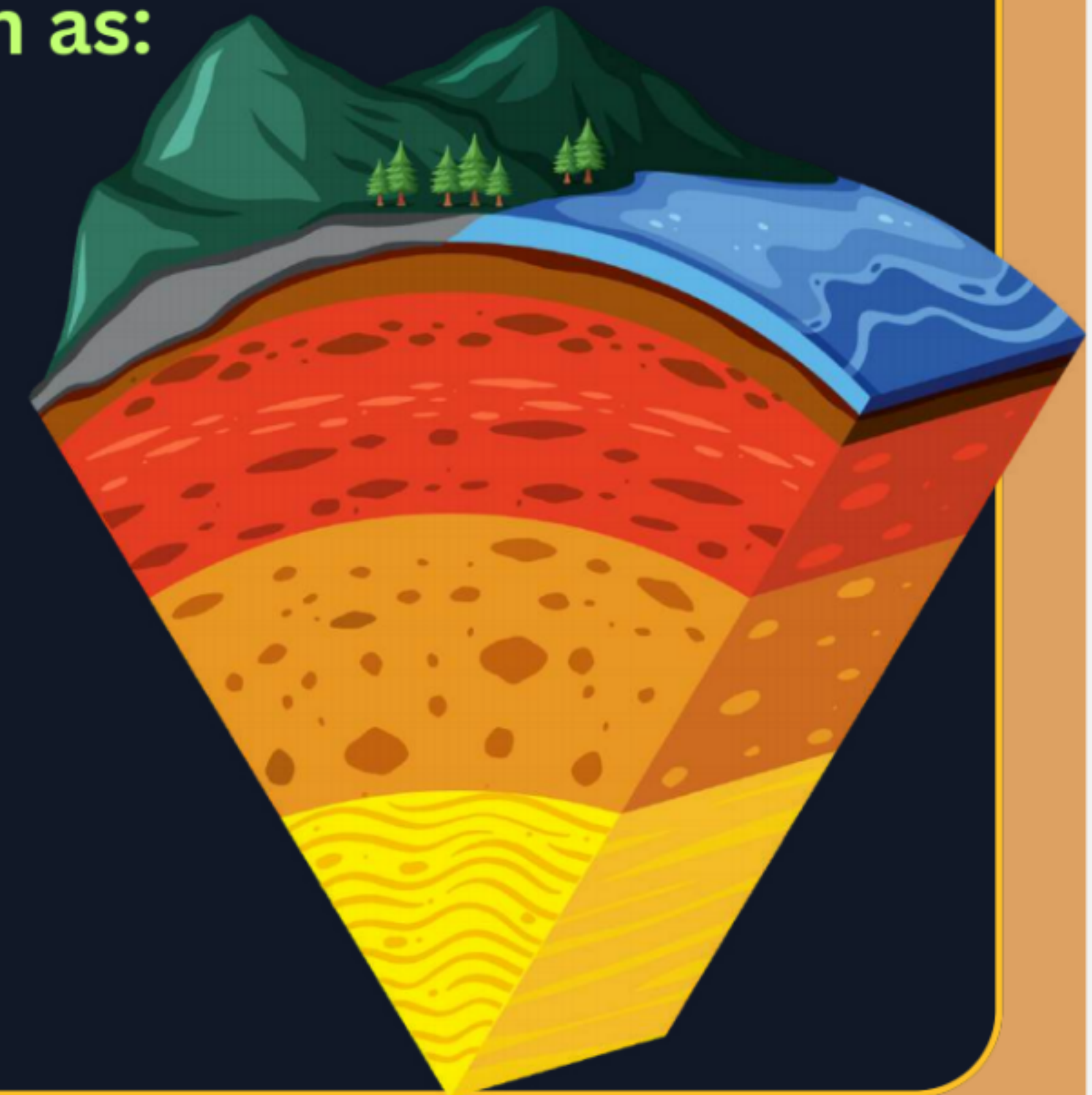
A composite image showing the Earth's internal layers. The background features a view of the Earth from space, with a semi-transparent cutaway revealing the interior. The layers shown are the crust (outermost, grey), the mantle (divided into upper and lower, with the lower mantle being red and glowing), and the core (divided into a liquid outer core and a solid inner core, both glowing red).

OCCURRENCE OF METALS

Sources of Metals

- Earth's crust is the major source of metals.
- Seawater also contains soluble metal salts such as:
 - Sodium chloride (NaCl)
 - Magnesium chloride (MgCl_2)

Why is gold found naturally, but sodium isn't?
the answer lies in **reactivity** of metals.



Occurrence of metals

- Metals occur in the Earth's crust in various forms. They are usually found in combined or free (native) states depending on their reactivity.
- *Sodium and potassium are highly reactive and are never found in free state in nature.*
- *Oxygen is the most abundant element in the Earth's crust.*
- *Aluminum is the most abundant metal in the Earth's crust and the third most abundant element after oxygen and silicon.*
- *In the universe, the most abundant element is Hydrogen, followed by Helium.*



Metals occur in nature in two forms:

Free/Native State

Metal occurs as the element itself.

Examples: Gold (Au), Silver (Ag),
Platinum (Pt), Copper (Cu)

Combined State

Metal occurs in the form of compounds.

Examples: Zinc (Zn), Iron (Fe), Lead (Pb) — mainly as oxides, sulphides or carbonates. Copper and silver may also occur as sulphide or oxide ores.

The form in which a metal occurs mainly depends upon its reactivity

Essential Terms

Metallurgy:

is the science and technology of extracting metals from their ores, refining them, and preparing them for use.



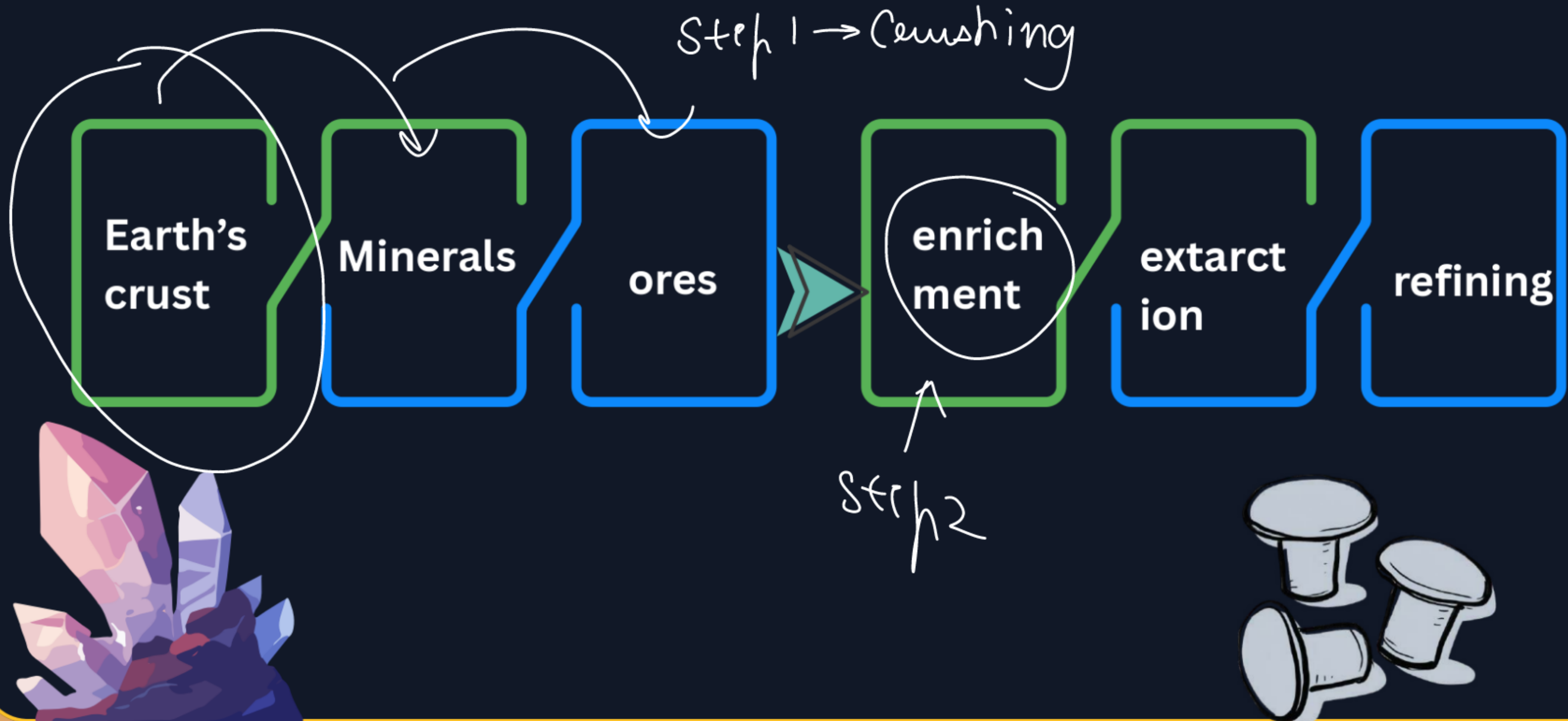
Gangue particles:

These are unwanted substances like sand, clay, oil, etc., found along with ores.

Concentration of ore:

the process by which gangue is removed to enrich the metal content in the ore.

Journey of Metal Extraction



Q. Define minerals, ores and gangue.

MINERALS

The elements or compounds which occur naturally in the Earth's crust are called minerals.



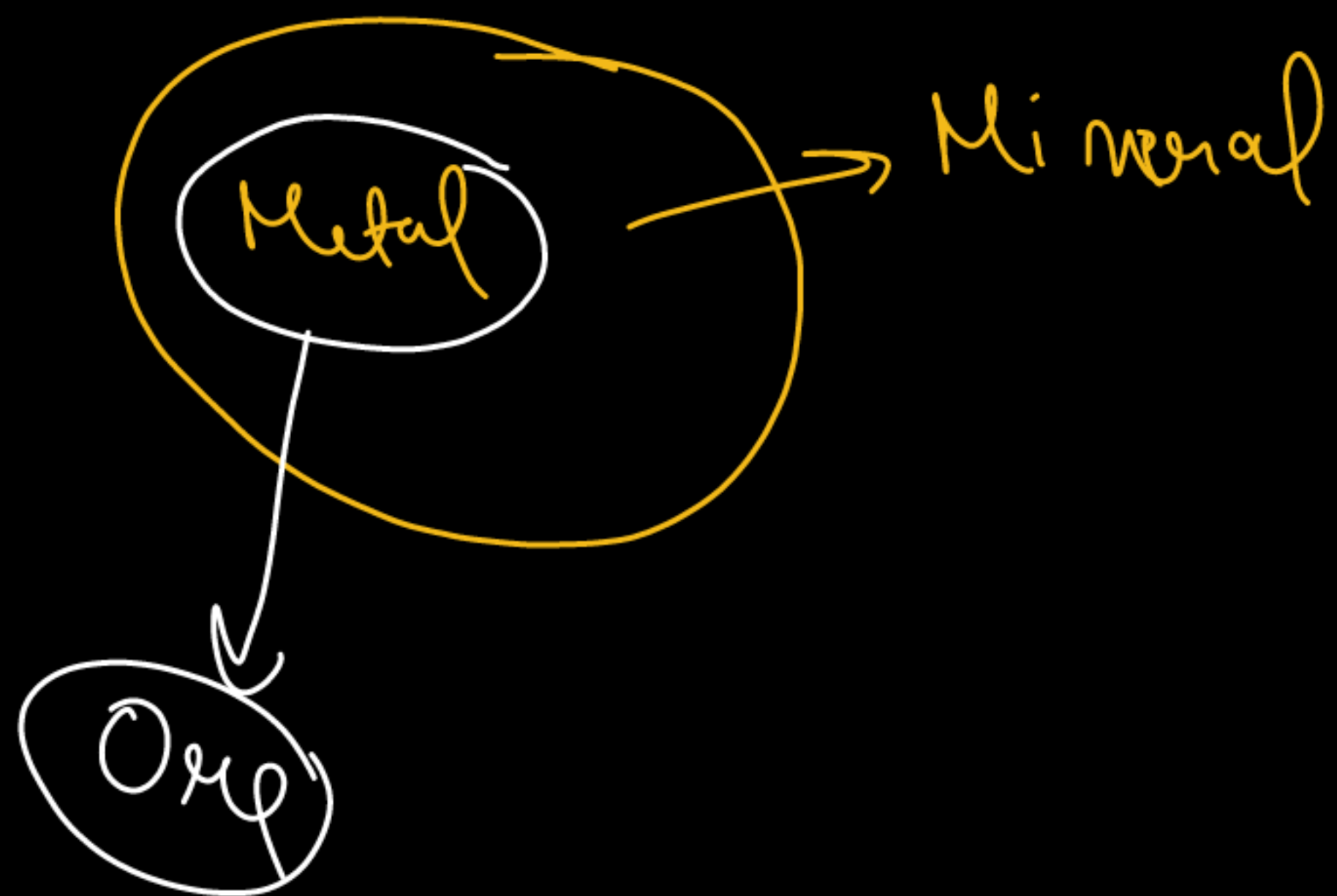
ORES

Minerals from which a metal can be extracted profitably are called ores.

Ore → contains sufficient amount of metal → metal can be extracted profitably



Hence, all ores are minerals, but all minerals are not ores.



Ores of Some Important Metals

Metal	Ore	Kind of Ore	Formula
Aluminium	→ Bauxite → Al_2O_3	Hydrated oxide ore	$Al_2O_3 \cdot 2H_2O$
Iron	Haematite	Oxide ore	→ Fe_2O_3
	Magnetite	Oxide ore	→ Fe_3O_4
Copper	✗ Copper pyrites	Sulphide ore	$CuFeS_2$
	Malachite	Carbonate ore	$CuCO_3 \cdot Cu(OH)_2$
	Cuprite	Oxide ore	Cu_2O
	Copper glance	Sulphide ore	→ Cu_2S
Zinc	Zinc blende / Sphalerite	Sulphide ore	ZnS
	Calamine	Carbonate ore	$ZnCO_3$
	✗ Zincite	Oxide ore	ZnO
Mercury	Cinnabar	Sulphide ore	HgS
Lead	✗ Galena	Sulphide ore	PbS

The ores of many metals are oxides, because oxygen is very reactive and abundant in the earth's crust.

K
Na
Ca
Mg
Al
Zn
Fe
Pb
H
Cu
Hg
Ag
Au
Pt

VERY REACTIVE METALS

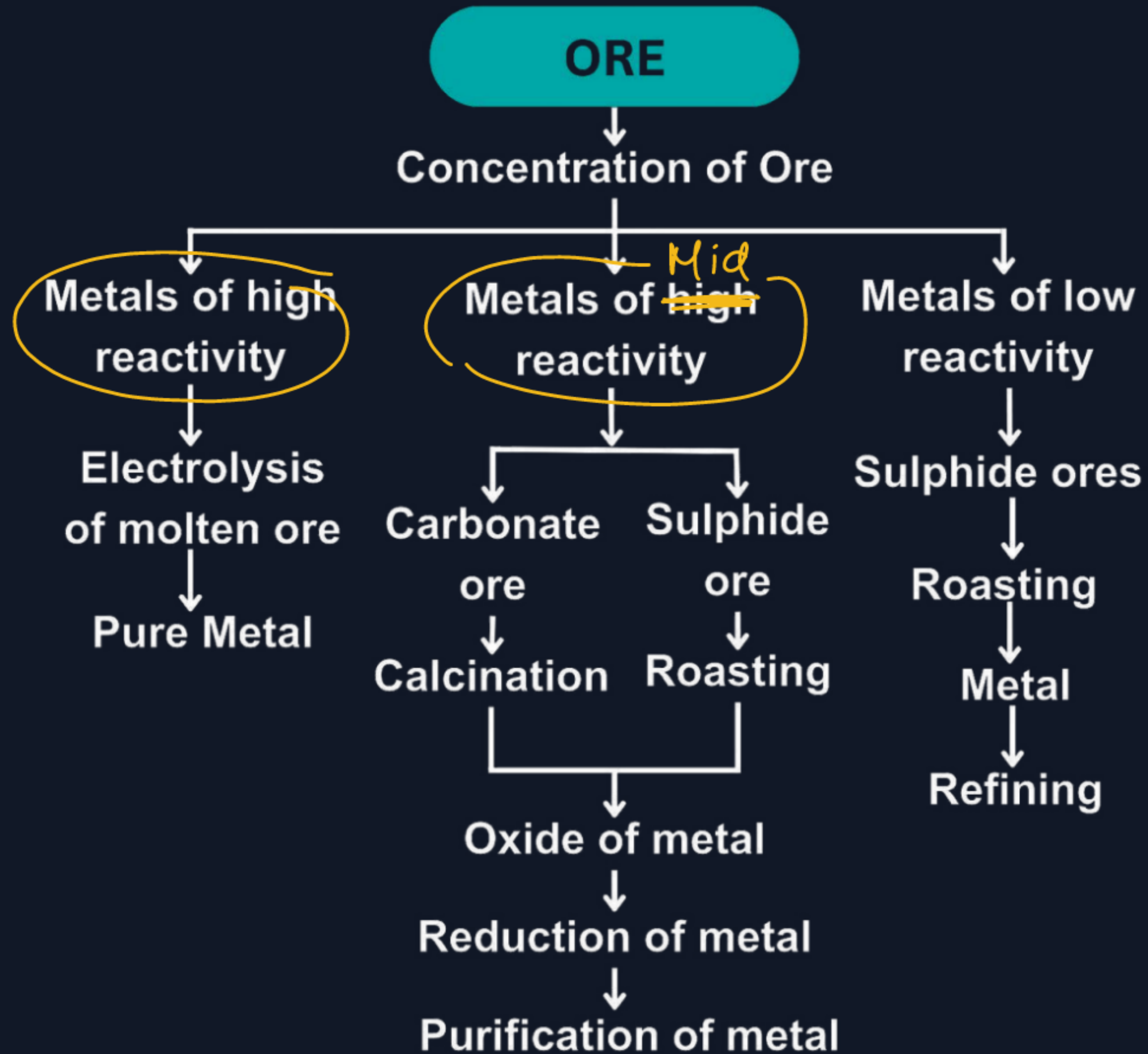
- They are never found in free state in nature.
- They always occur in combined state as their oxides, carbonates or sulphides.

MODERATELY REACTIVE METALS

- They are found in the Earth's crust mainly as oxides, sulphides and carbonates. Zinc → as ZnS (zinc blende), ZnCO₃ (calamine), Iron → as Fe₂O₃ (haematite), Lead → as PbS (galena)

LEAST REACTIVE METALS.

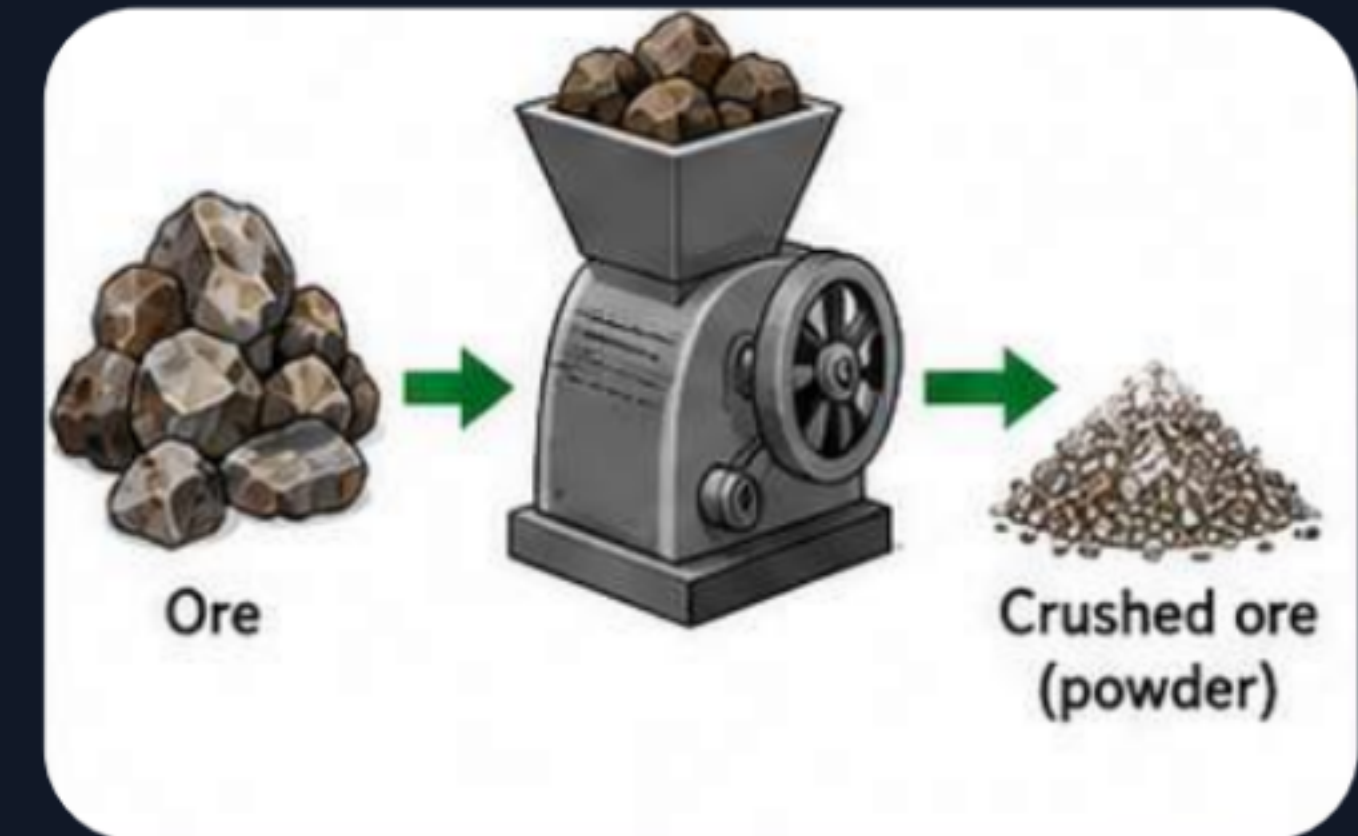
- They are found in free state (native state) as well as in combined state.
- Examples: Copper, Silver, Mercury → found both free and combined (as oxides/sulphides).
- Gold, Platinum → found in free (native) state.



Steps in Metallurgy

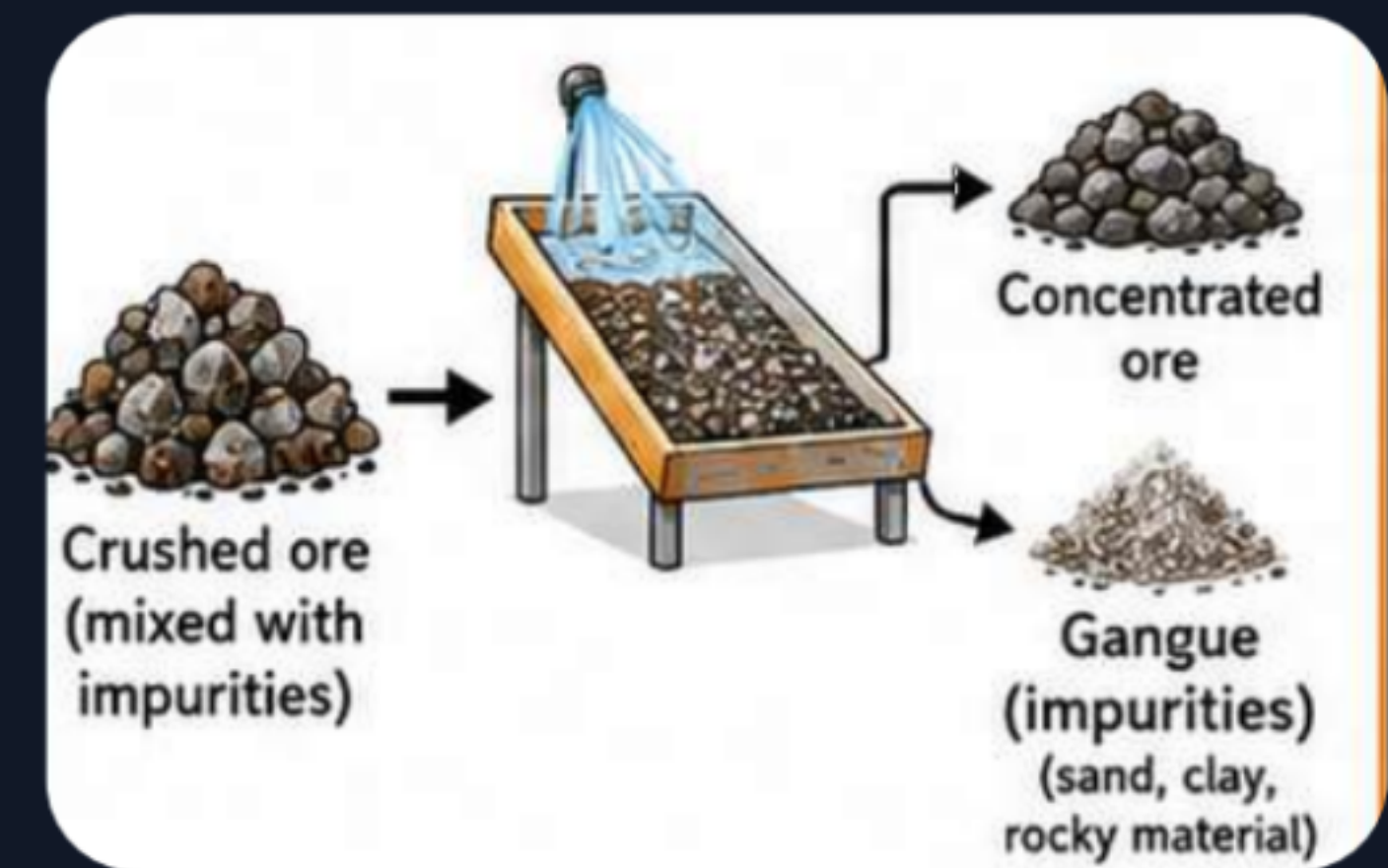
✓ Step 1: Crushing and Grinding of Ore

- First step of the metallurgical process.
- Ore is crushed and ground into smaller pieces/powder.
- Purpose → makes extraction of metal easier and more efficient.



✓ Step 2: Concentration of Ore (Enrichment)

- Process of removing gangue (impurities) like sand, clay, and rocky materials from ore.
- Increases the percentage of metal in the ore.
- Done by various physical/chemical methods depending on the ore and impurity.



Step 3: Extraction of Metals

- Process of obtaining metal from concentrated ore.
- Achieved by reduction (removal of oxygen or non-metallic element).
- Method depends on the reactivity of the metal:
- **High reactivity metals (K, Na, Ca, Mg, Al):** extracted by electrolysis.
- **Medium reactivity metals (Zn, Fe, Pb, etc.):** extracted by reduction with carbon/CO, or displacement.
- **Low reactivity metals (Au, Ag, Cu, etc.):** found in native state or extracted by mild reducing agents.



1 High reactivity metals
K, Na, Ca, Mg, Al



3 Low reactivity metals
Au, Ag, Cu, etc.



2 Medium reactivity metals
Zn, Fe, Pb, etc.



Check yourself

QUES. A mining company discovers three minerals containing the same metal

Mineral	Metal content	Cost of Extraction
P	high	low
Q	high	very high
R	low	low

The company decides to extract the metal only from mineral P

1. Which mineral is most likely to be called an ore?
2. Why are minerals Q and R not preferred for commercial extraction?

Check yourself

QUES: Assertion–Reason (Board Pattern)

Assertion (A)

A mining company selects Bauxite for commercial extraction instead of another aluminum-containing mineral. T

Reason (R)

Only those minerals from which metals can be extracted conveniently and economically are called ores. T

- ✓ A. Both A and R are true, and R is the correct explanation.
- B. Both are true, but R is not an explanation.
- C. True, R false.
- D is false, and R is true.



Check yourself

QUES: A geologist discovers two metals underground.

- Metal X is found in the free state. *low*
- Metal Y is always found combined with oxygen. *Reactivity*

Explain what these observations indicate about the ~~relativity~~ of X and Y.

Ans: Metal X is found in the free (native) state, indicating that it is less reactive. Such metals do not react easily with air, water, or other substances and can occur in nature in the free state. (Example: Gold, Platinum)

Metal Y is always found combined with oxygen, indicating that it is more reactive. It readily reacts with oxygen to form metal oxides and is therefore found in the combined state. (Example: Aluminium, Magnesium)



Check yourself

QUES: If metals occur naturally, all should be found in free state.” Do you agree?

Answer: No, I do not agree with the statement.

- According to the reactivity series, only very unreactive metals (such as gold, silver, and platinum) are found in the free (native) state because they do not react easily with air, water, or other substances.
- Most metals are reactive and readily combine with oxygen, sulphur, carbonates, etc., to form ores. Hence, they occur in the combined state.



Check yourself

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QUES: Three minerals are found in a mining area: Mineral A: Fe_2O_3 , Mineral B: $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ and Mineral C: SiO_2

The mining company wants to extract metals economically.

Answer the following:

- (a) Which of the above are ores?
- (b) Which metal is obtained from Mineral A?
- (c) Why is Mineral C not considered an ore?

Answer

- (a) Mineral A (Haematite) and Mineral B (Bauxite) are ores.
- (b) Iron is extracted from Mineral A.
- (c) Mineral C is silica (SiO_2). It does not contain a metal that can be extracted economically, so it is not an ore.



**My friends are fighting over
whether the answer is
'iron ore' or 'aluminium ore'**



**me who wrote
'Teri Ore' in exams**